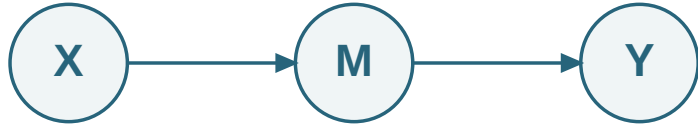


Graph surgery for a node intervention

Observational system



$$M = f_M(X, U_M)$$

$$Y = f_Y(M, U_Y)$$

Intervened system: $\text{do}(M = m)$



removed

$$M = m$$

$$Y = f_Y(M, U_Y) \text{ (unchanged)}$$

Replace only the equation for M with a constant.

Remove $X \rightarrow M$; keep $M \rightarrow Y$ and propagate the intervention downstream.